

Exercise session 02

Introduction to C++.

Advanced Programming - SISSA, UniTS, 2025-2026

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Exercise 1: control structures

Write a C++ program `temperature_converter` that converts a given temperature from Celsius to Fahrenheit or viceversa.

- The program takes an input number from the user and the input unit as a string (or character).
- Use conditional statements to check if the user has provided a temperature in Fahrenheit or Celsius and print the corresponding output.

- The conversion formulas are:

$$T_{\text{Fahrenheit}} = \frac{9}{5}T_{\text{Celsius}} + 32.$$

$$T_{\text{Celsius}} = \frac{5}{9}(T_{\text{Fahrenheit}} - 32).$$

Exercise 2: memory management

Create a C++ program that dynamically allocates memory for an array of integers.

- Allow the user to specify the size of the array.
- Fill the array with random integers.
- Write a function to find and display the maximum and minimum values in the array. **Hint:** the maximum and minimum value are stored in two variables passed as references to this function.

Exercise 3: complete the missing statistics calculator

The `hints/ex3/` folder provides a partially implemented C++ program for calculating and displaying statistics for a set of numbers. Your task is to fill in the missing parts to make the program functional.

- Use the provided function prototypes in `statistics.hpp` to guide your implementation in `statistics.cpp`.
- You may need to sort the input numbers to calculate the median. You can use standard C++ functions like `std::sort` for this purpose.
- Implement a loop in `main.cpp` to allow the user to calculate statistics for multiple sets of numbers without exiting the program.

Exercise 4: `struct`

- Define a `struct` called `Student` that represents information about a student, including their name, age, and grade average.
- Create a `std::vector` of 5 `Student` objects and initialize them with sample data.
- Write a function to display the information of all students in the array.
- Organize your code by separating the struct definition, data initialization, and display function into different files or modules.

Exercise 5: code organization

Write a C++ program that simulates a simple calculator. Define functions for addition, subtraction, multiplication, and division. Allow the user to enter two numbers and choose an operation. Perform the chosen operation and display the result.

Here's a breakdown of the project structure:

- `calculator/`
 - `src/`
 - `main.cpp`
 - `calculator.cpp`
 - `include/`
 - `calculator.hpp`
 - `build/` (build artifacts, such as object files and executables).
 - `build.sh` (a Bash script that compiles and properly links the code)